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Course Name:5G 6G Communication Systems

Course Code:ECA5601

Ex:12: NB-IoT vs LTE-M Throughput Simulation Date

Aim:

To simulate and compare the performance of NB-IoT and LTE-M communication technologies in terms of Throughput (Mbps), Latency (ms), and Packet Delivery Ratio (%) using MATLAB.

Procedure

1. Define system parameters for NB-IoT and LTE-M such as bandwidth, transmission power, and modulation efficiency.

2.Generate network conditions by varying Signal-to-Noise Ratio (SNR) or channel quality values.

3.Compute Throughput (Mbps) for both NB-IoT and LTE-M using their respective data rate models.

4.Calculate Latency (ms) based on transmission time, scheduling delay, and network load.

5.Determine Packet Delivery Ratio (PDR %) by comparing successfully received packets with transmitted packets.

6.Simulate multiple iterations to observe performance under different channel conditions.

7.Plot comparison graphs for Throughput, Latency, and PDR between NB-IoT and LTE□M.

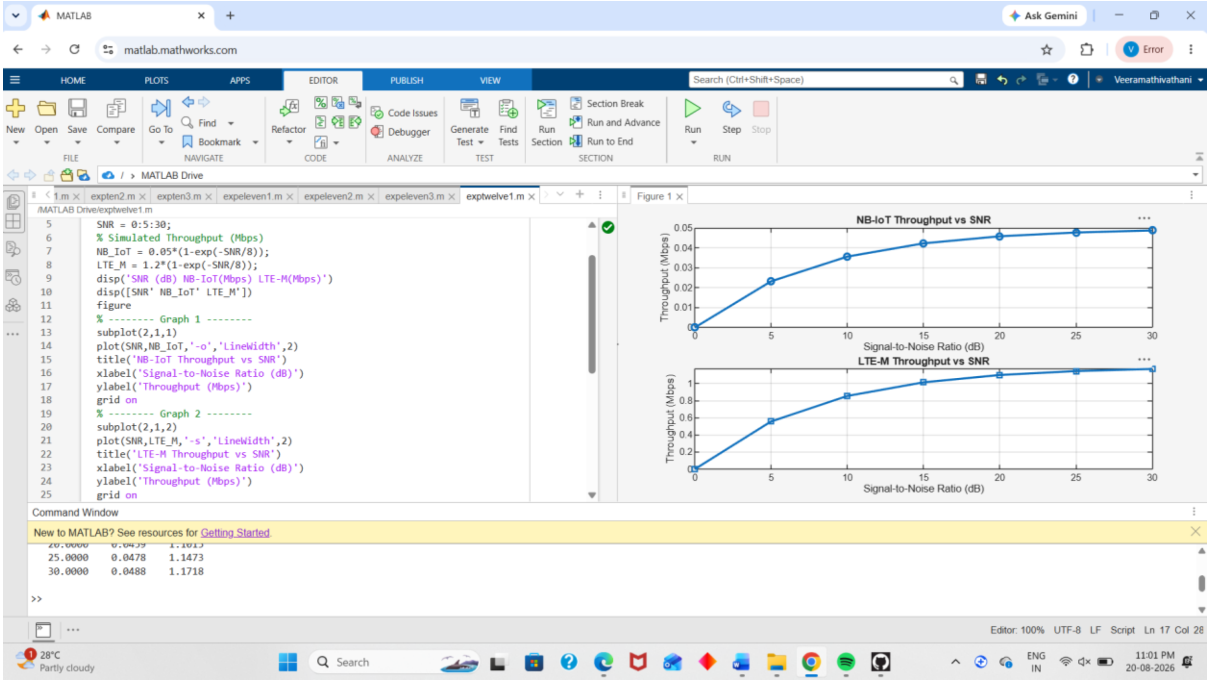
8.Analyze results to identify which technology performs better under different IoT scenarios.

Objective 1: Matlab Code and Output

```
clc;

clear;
```

```
close all;
% SNR values (dB)
SNR = 0:5:30;
% Simulated Throughput (Mbps)
NB_IoT = 0.05*(1-exp(-SNR/8));
LTE_M = 1.2*(1-exp(-SNR/8));
disp('SNR (dB) NB-IoT(Mbps) LTE-M(Mbps)')
disp([SNR' NB_IoT' LTE_M'])
figure
% -------- Graph 1 --------
subplot(2,1,1)
plot(SNR,NB_IoT,'-o','LineWidth',2)
title('NB-IoT Throughput vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Throughput (Mbps)')
grid on
% -------- Graph 2 --------
subplot(2,1,2)
plot(SNR,LTE_M,'-s','LineWidth',2)
title('LTE-M Throughput vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Throughput (Mbps)')
grid on
```



Objective 2: Matlab Code and Output

clc;

clear;

close all;

% SNR values (dB)

SNR = 0:5:30;

% Simulated Latency (ms)

NB_IoT = [180 160 145 130 120 110 100];

LTE_M = [90 80 70 60 50 45 40];

disp('SNR(dB) NB-IoT Latency(ms) LTE-M Latency(ms)')

disp([SNR' NB_IoT' LTE_M'])

figure

% -------- Graph 1 --------

subplot(2,1,1)

plot(SNR,NB_IoT,'-o','LineWidth',2)

title('NB-IoT Latency vs SNR')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Latency (ms)')

grid on

% -------- Graph 2 --------

subplot(2,1,2)

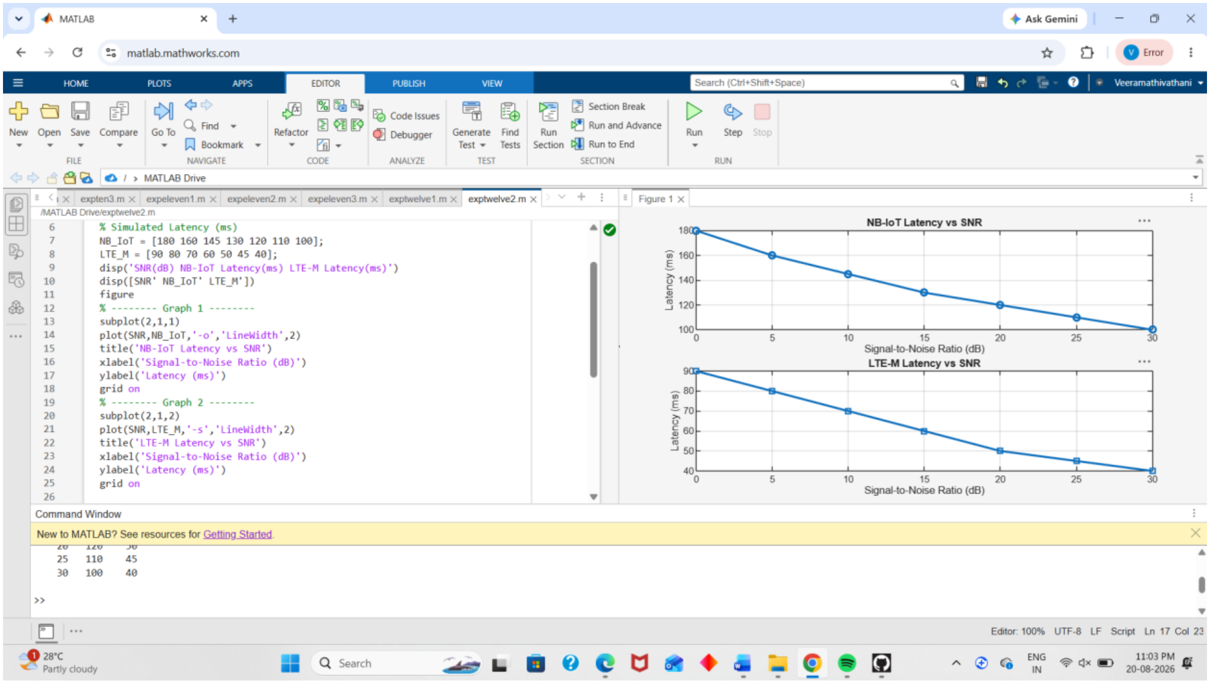
plot(SNR,LTE_M,'-s','LineWidth',2)

title('LTE-M Latency vs SNR')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Latency (ms)')

grid on



Objective 3: Matlab Code and Output

clc;

clear;

close all;

% SNR values (dB)

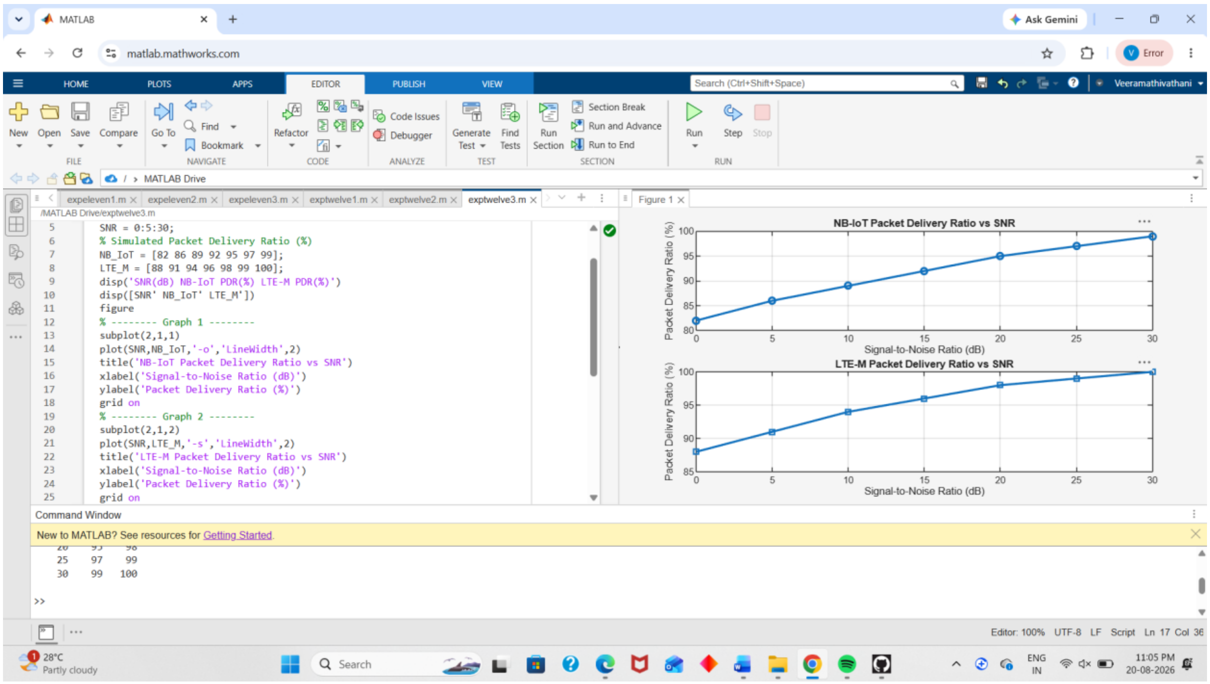
SNR = 0:5:30;

% Simulated Packet Delivery Ratio (%)

NB_IoT = [82 86 89 92 95 97 99];

LTE_M = [88 91 94 96 98 99 100];

```
disp('SNR(dB) NB-IoT PDR(%) LTE-M PDR(%)')
disp([SNR' NB_IoT' LTE_M'])
figure
% -------- Graph 1 --------
subplot(2,1,1)
plot(SNR,NB_IoT,'-o','LineWidth',2)
title('NB-IoT Packet Delivery Ratio vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Packet Delivery Ratio (%)')
grid on
% -------- Graph 2 --------
subplot(2,1,2)
plot(SNR,LTE_M,'-s','LineWidth',2)
title('LTE-M Packet Delivery Ratio vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Packet Delivery Ratio (%)')
grid on
```



Result:

1. The throughput performance of both NB-IoT and LTE-M was successfully simulated, and LTE-M achieved higher throughput than NB-IoT under the same network conditions.

2.The latency analysis showed that LTE-M provides lower communication delay compared to NB-IoT, making it more suitable for real-time IoT applications.

3.The Packet Delivery Ratio (PDR) of both technologies improved with increasing SNR, with LTE-M demonstrating slightly higher reliability than NB-IoT during data transmission